

S. Zach Bagley
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EDUCATION

Master of Science, Information Systems — Emphasis in Software Engineering Provo, UT
Bachelor of Science, Information Systems Aug 2021 – Apr 2026
Brigham Young University – Marriott School of Business

- GPA 3.85
- ACT 33 (98th percentile)
- Association of Information Systems
- *BYU Marriott School Dean's List*

EXPERIENCE

Software Engineering Intern Idaho Falls, ID
Melaleuca, Inc. May 2025 – Aug 2025

- Developed a modular system for retrieving profile data across 20 C#/.NET APIs and several Microsoft SQL Server databases, reducing API and database calls by up to 80% in 75% of cases, improving performance overall performance
- Designed and implemented partial profile update and caching using Kafka consumers, enabling targeted cache refreshes to lighten processing load
- Built a C# console app and supporting API routes to streamline PDF generation and storage of customer enrollment data; utilized application in automated jobs with JAMS
- Engineered robust AI workflows with Claude Code and Copilot, including implementation of custom skills and comprehensive guiding markdown files

Software Engineer Provo, UT
BYU J. Reuben Clark Law School Apr 2024 – May 2025

- Led backend development team for law.byu.edu and other BYU Law sites, including full-site modernization overhaul and implementation of security measures including SSO and MFA
- Configured and maintained secure cloud architecture for law.byu.edu and cpcr.byu.edu (AWS: ECS, Lambda)
- Enhanced security, functionality, and maintainability of BYU Law's internal information system's Node.js backend and MySQL database through refactoring of code architecture
- Implemented Next.js into law.byu.edu's React frontend

Software Engineering & Data Communications Teaching Assistant Provo, UT
BYU Information Systems Sep 2025 – Apr 2026

- Contributed to curriculum development (including training students on effective AI-augmented engineering with MCP, markdown files, etc.) on topics related to data TCP/IP networking, cloud infrastructure, and application development (particularly with C#/ASP .NET & TypeScript/React)
- Evaluated projects including code repositories, AWS infrastructure stacks, and full-stack development projects

IT Specialist Provo, UT
BYU Department of Continuing Education Sep 2023 – Apr 2024

- Managed system privileges, network configuration, and access control for DCE's appx. 800 employees, including compliance with organizational security policies
- Resolved technical issues to support secure and efficient IT environment
- Configured and maintained computer workstations

SKILLS & PROJECTS

- **Programming Languages** (ordered by proficiency): C#, JavaScript, TypeScript, SQL, Swift, Python, PHP, Java
- **AI-Augmented Development:** Built and maintained structured AI workflows to increase development speed and ensure consistent output quality, including at Melaleuca and in personal projects
- **Frameworks, Libraries, Tools, etc.:** .NET Core, Express (Node.js), SwiftUI, React, relational databases (MySQL, SQL Server, PostgreSQL), Vapor, Laravel, Vue, MongoDB, Snowflake, DBT
- **Portfolio:** Full-stack projects including alternative social media platform (Realspace; C#/.NET, Swift) and AI storyboarding app (Atelier; TypeScript/Node.js, React). See these projects and more at zachbagley.net/projects
- **Cloud & DevOps:** AWS (EC2, ECS, RDS, CloudFormation, S3, Lambda); Docker; CD/CI (CodePipeline); Git/GitHub